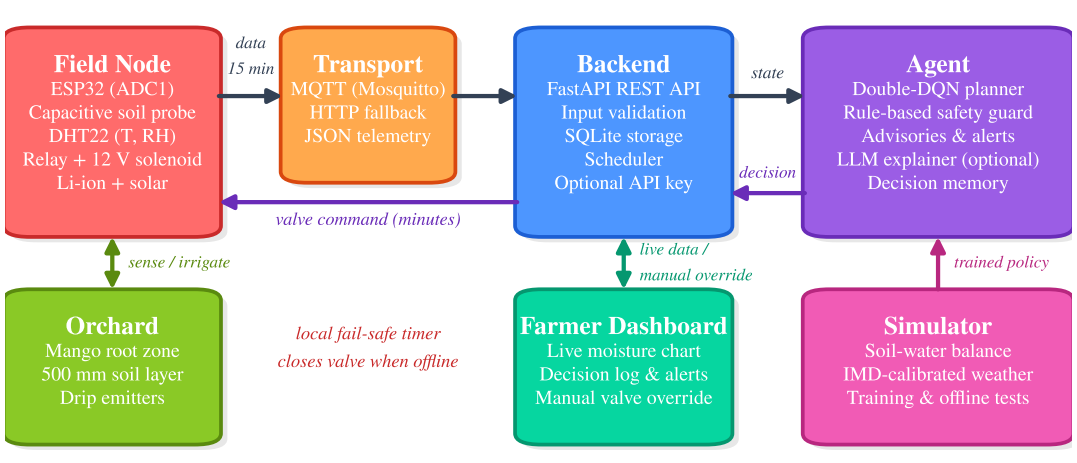
Field Node
ESP32 (ADC1)
Capacitive soil probe
DHT22 (T, RH)
Relay + 12 V solenoid
Li-ion + solar
data
15 min
Transport
MQTT (Mosquitto)
HTTP fallback
JSON telemetry
Backend
FastAPI REST API
Input validation
SQLite storage
Scheduler
Optional API key
state
Agent
Double-DQN planner
Rule-based safety guard
Advisories & alerts
LLM explainer (optional)
Decision memory
decision
valve command (minutes)
sense / irrigate
live data /
manual override
trained policy
Orchard
Mango root zone
500 mm soil layer
Drip emitters
local fail-safe timer
closes valve when offline
Farmer Dashboard
Live moisture chart
Decision log & alerts
Manual valve override
Simulator
Soil-water balance
IMD-calibrated weather
Training & offline tests